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Brief Correspondence

Comparison of Typical Prostate Adenocarcinoma and Rare Histological Variant Prostate Cancer Showed Different Characteristics and Prognosis: A Surveillance, Epidemiology, and End Results Database Analysis

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Abstract

There are currently few large-scale studies describing the clinical characteristics of these histological variants of prostate cancer (PCa) and the prognostic results of these patients. Thus, we used the Surveillance, Epidemiology, and End Results (SEER) database to achieve a more comprehensive understanding of the current prevalence and prognosis of these histological variants. A total of 780 (3.3%) patients with histological variants and 235 834 patients with typical adenocarcinoma were included in the present study. The variants group presented with a more advanced stage than the typical adenocarcinoma group. The results also showed that patients with neuroendocrine (NE) PCa were more likely to have visceral metastases. In prognostic comparison, we found that variants, except for mucinous, had a worse prognosis than typical adenocarcinoma. Moreover, among these variants, we also discovered that NEPCa patients with low prostate-specific antigen level had the worst survival. In contrast, mucinous PCa had similar overall survival to typical adenocarcinoma.

Patient summary: We reported follow-up results for prostate cancer (PCa) patients from the Surveillance, Epidemiology, and End Results database (2004–2016). We focused on the prognosis of PCa with different histological variants and evaluated the risk factors that were significant in predicting poor prognosis. We suggested that neuroendocrine PCa patients should receive active treatment as soon as possible to improve their prognosis. We still need further studies to identify the role of mucinous PCa in patients' prognosis compared with typical adenocarcinoma.

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Prostate cancer (PCa) is the most common malignant tumor in men, and there were >240 000 new cases of PCa and >30 000 deaths caused by PCa in the USA in 2020 [1]. Most of the pathological types of PCa are acinar adenocarcinoma. However, some unusual histological variants account for 5–10% of PCa [2]. Few clinical studies had analyzed the prog-

nosis of these variants comparatively and comprehensively. Therefore, we included patients with typical adenocarcinoma (ICD-O-3 code: 8140/3), mucinous PCa (ICD-O-3 codes: 8480/3, 8481/3), signet ring cell (SRC) carcinoma (ICD-O-3 8490/3), ductal PCa (ICD-O-3: 8500/3, 8201/3, 8260/3, 8230/3), or neuroendocrine (NE) PCa (ICD-O-3

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8013/3, 8041/3, 8246/3) from 2004 to 2016 in the Surveillance, Epidemiology, and End Results (SEER) database [3]. The diagnosis was confirmed by histological examination from either biopsy or surgical pathology. Patients without pathological diagnosis and survival-related data were excluded (Supplementary Fig. 3). Overall survival (OS) is the time from diagnosis to death due to any cause, with censoring at the last visit date within SEER or the date of death due to any cause. Cancer-specific survival (CSS) is the time from diagnosis to death from PCa, with censoring at the last visit date within SEER or the date of death due to any other cause. Our statistical analysis included Pearson's chi-square tests of categorical variables, and Kaplan-Meier curves and the log-rank test were used to assess survival outcomes stratified by divergent groups (including typical adenocarcinoma vs all variants, acinar vs nonacinar variants, and each variant vs typical adenocarcinoma). A multivariable Cox regression analysis was used to identify prognostic variables in different groups.

Table 1 – Clinical parameters of patients diagnosed with typical adenocarcinoma and all variants between 2004 and 2016

Variables	Typical adenocarcinoma (n = 235 834)	All variants of PCa (n = 780)	p value
Age at diagnosis (yr; %)			<0.001
18–59	24	22	
60–74	58	50	
≥75	18	28	
Race (%)			0.4
White	78	78	
PSA (ng/ml; %)			<0.001
0–4.0	5.6	11	
4.1–10.0	28	21	
10.1–20.0	6.4	7.0	
>20	6.0	13	
Unknown	54	48	
Tumor grade (%)			<0.001
Low	48	12	
High	49	68	
Unknown	3.0	20	
Gleason score (%)			<0.001
≤6	19	3.0	
3 + 4/4 + 3	21	11	
≥8 or primary = 5	10.0	27	
Unknown	50	59	
Summary stage (%)			<0.001
Localized	81	45	
Regional	13	27	
Distant	4.4	27	
Unknown	1.6	1.0	
T stage (%)			<0.001
T1/T2	81	51	
T3/T4	10.0	29	
Unknown	9.0	20	
N stage (%)			<0.001
N0	86	65	
N1	2.0	14	
Unknown	12	21	
M stage (%)			<0.001
M0	86	62	
M1	4.0	22	
Unknown	11	16	
Surgery (%)			<0.001
Yes	37	52	

PCa = prostate cancer; PSA = prostate-specific antigen.

A total of 236 614 PCa patients were included (Table 1), 235 843 (97%) were typical carcinoma and 780 (3.3%) atypical variants. There were 30% NE, 49% ductal, 4.0% SRC, and 17% mucinous variants (Supplementary Fig. 1A). Case characterization and clinical features in each variant are presented in Supplementary Table 1. Patients with NEPCa had a higher risk of visceral metastases than other variants (Supplementary Fig. 1B).

Kaplan-Meier curves revealed that all variants had worse CSS and OS than typical adenocarcinoma (Fig. 1A). The multivariable analysis also presented that all variants had a higher risk of poor outcomes than typical adenocarcinoma (CSS: hazard ratio [HR] = 2.43, OS: HR = 1.85, all $p < 0.001$; Supplementary Table 2). Patients who underwent surgery had longer CSS and OS (CSS: HR = 0.53, OS: HR = 0.68, all $p < 0.001$; Supplementary Table 2). Age, prostate-specific antigen (PSA), T stage, and N stage had an independent influence on OS (Supplementary Fig. 2). In addition, we divided all variants into acinar and nonacinar carcinoma groups. Acinar carcinoma variants (mucinous and SRC) had better survival than the nonacinar group (ductal and NE) in terms of CSS and OS (all $p < 0.001$) in Kaplan-Meier curves (Fig. 1B). Nonacinar carcinoma variants were significantly associated with worse CSS and OS than acinar carcinoma variants (CSS: HR = 4.09, OS: HR = 1.83, all $p < 0.001$; Supplementary Table 3). In addition, patients with high tumor grade had worse CSS and OS than those with low tumor grade (CSS: HR = 3.09, OS: HR = 1.63, all $p < 0.05$; Supplementary Table 3).

The multivariable analysis showed that ductal, NE, and SRC variants were associated with worse prognosis including OS and CSS, than typical adenocarcinoma (ductal—CSS: HR = 1.45, OS: HR = 1.25; NE—CSS: HR = 4.23, OS: HR = 3.90; SRC—CSS: HR = 2.39, OS: HR = 1.97, all $p < 0.05$; Supplementary Table 4). Zaffuto et al [4] reported similar conclusion based on the SEER database (2004–2013) and believed that the prognosis of NEPCa was poor regardless of histological type, especially in the metastatic stage. NEPCa lost androgen receptor (AR) expression, which led to the loss of sensitivity to endocrine therapy. Patients with mucinous PCa had better CSS than typical adenocarcinoma (HR = 0.37, $p = 0.027$; Supplementary Table 4). Some studies suggested that mucinous PCa had better survival [5,6]. However, others identified that mucinous carcinoma had poor prognosis and more advanced stage [7]. In our study, NEPCa had the worst prognosis among all variants in CSS and OS (Fig. 1C). The multivariable analysis also indicated that NEPCa had the worst survival (CSS: HR = 4.23, OS: HR = 3.90, all $p < 0.05$). Wang et al [8] discussed that lower PSA levels were associated with worse prognosis in patients with either NEPCa or NE differentiation. We also found that NEPCa patients with high tumor grade had worse OS than those with low tumor grade (OS: HR = 3.74, $p = 0.029$; Supplementary Table 5). Although these histological variants were rare in a clinic, some variants still needed to be discussed and analyzed.

Marcus et al [9] discussed the incidence and survival in patients with rare variants of PCa registered in the SEER database (1973–2008). There are significant differences in the incidence and survival between Caucasian and African

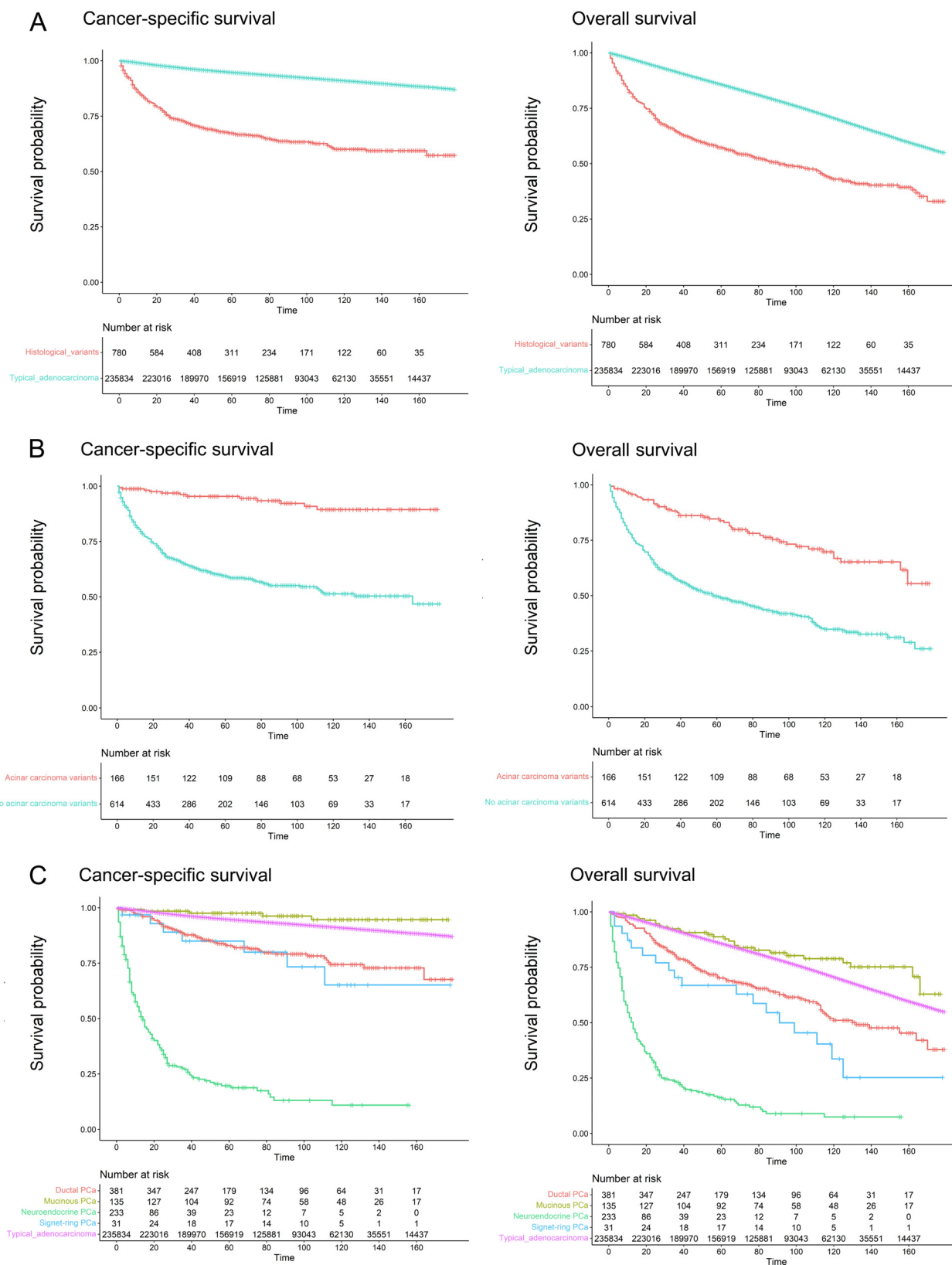


Fig. 1 – Cancer-specific and overall survival based on different histological subgroups. (A) Cancer-specific and overall survival of patients with all variants and typical adenocarcinoma. (B) Cancer-specific and overall survival of patients with acinar and nonacinar carcinoma variants. (C) Cancer-specific and overall survival of patients with different histological variants. PCa = prostate cancer.

American patients with mucinous carcinoma. However, our study did not find that race had a statistical impact on the

prognosis of mucinous PCa patients. Bronkema et al [10] suggested that NE, adenosquamous, sarcomatoid, SRC, and

ductal variants had a higher risk of metastases than typical adenocarcinoma. NEPCa was associated with the worst OS and mucinous PCa with the best OS. However, we did not find mucinous PCa to have longer OS than typical adenocarcinoma.

We admitted that due to the limitations of this study, we could not evaluate the pathology in the SEER database directly, and patients in SEER could receive inconsistent treatments in different institutions.

To summarize, we conducted an up-to-date detailed analysis using the SEER database, suggesting that patients with NEPCa have poor prognosis and need active treatment, while the prognosis of patients with mucinous PCa remains controversial and needs further study.

Author contributions: Jiatong Zhou had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Zhou.

Acquisition of data: Zhou.

Analysis and interpretation of data: Zhou, Ding.

Drafting of the manuscript: Zhou.

Critical revision of the manuscript for important intellectual content: Zhou, Ding.

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Appendix A – Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eururo.2022.01.028>.

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